Table 1: Primary family parameters and attributed diagnostics

Scheme	M/u/K (bits)	CT (bytes)	PK (bytes)	CT+PK (bytes)	C diag.	Q diag.	b ≈
S512-q251	256/256/256	543	511	1054	140.05	134.56	166.15
S1024-q347	256/512/256	1113	1081	2194	270.07	251.59	163.38
D512-q157	128/128/256	483	467	950	144.39	138.45	133.38
D1024-q251	256/256/256	1053	1021	2074	278.68	259.33	166.86

M/u/K are external message / internal residue / shared-key output lengths in bits.
C/Q are supplied structural GSA/MATZOV diagnostics; b is approximate -log2(B), not a rounded strict inequality.
These columns are not a full KEM security claim or a NIST certificate. No compression denominator is assigned.

Table 2: Primary profiles and matched peers: local KEM performance

	REF (kcycles)			AVX2 (kcycles)		
Scheme	KeyGen	Encaps	Decaps	KeyGen	Encaps	Decaps
BAT-512	19,668.915 [14,325.761, 24,553.788]	47.020 [46.978, 47.112]	300.402 [300.285, 300.454]	17,832.227 [16,642.649, 21,404.566]	10.061 [10.050, 10.070]	56.612 [56.586, 56.660]
END-512 [1]	9,216.122 [8,418.315, 10,016.721]	28.908 [28.796, 28.988]	328.230 [328.078, 328.425]			
S512-q251	2,222.412 [2,220.931, 2,229.777]	300.676 [300.543, 300.744]	460.114 [459.999, 460.254]	1,843.785 [1,841.390, 1,845.584]	184.417 [184.247, 184.541]	232.698 [232.421, 232.929]
D512-q157	2,260.666 [2,257.419, 2,264.095]	268.340 [268.166, 268.436]	427.895 [427.696, 428.057]	1,878.953 [1,875.310, 1,882.568]	152.781 [152.730, 152.863]	201.447 [201.392, 201.552]
BAT-1024	106,227.522 [77,951.843, 135,044.927]	93.911 [93.849, 93.950]	621.914 [621.791, 622.012]	97,193.241 [83,738.468, 104,319.250]	21.241 [21.225, 21.269]	116.493 [116.439, 116.555]
END-1024 [2,3]						
S1024-q347	4,917.572 [4,913.513, 4,920.746]	741.887 [741.716, 741.972]	1,065.250 [1,065.098, 1,065.653]	3,861.338 [3,859.300, 3,865.328]	484.334 [484.092, 484.443]	560.771 [560.662, 560.828]
D1024-q251	4,648.099 [4,642.902, 4,657.134]	696.215 [696.013, 696.385]	1,023.468 [1,023.243, 1,023.748]	3,568.026 [3,566.258, 3,571.190]	441.248 [441.100, 441.425]	521.723 [521.609, 521.894]

Median of accepted batch means; brackets give the prespecified bootstrap 95% interval.
1 kcycle = 1000 raw hardware CPU cycles. Counter-read overhead is retained, not subtracted.
Blank performance cells are correctness exclusions, never zero costs. Batch counts are retained in CSV.
[1] END-512 / AVX2: Shared-key preflight failed through the preserved-source adapter; no timing is ranked
[2] END-1024 / REF: Shared-key preflight failed through the preserved-source adapter; no timing is ranked
[3] END-1024 / AVX2: Shared-key preflight failed through the preserved-source adapter; no timing is ranked

Table 3: Appendix: all16 selected family profiles

Scheme	M/u/K (bits)	CT (bytes)	PK (bytes)	CT+PK (bytes)	C diag.	Q diag.	b ≈
S512-q251	256/256/256	543	511	1054	140.05	134.56	166.15
S1024-q347	256/512/256	1113	1081	2194	270.07	251.59	163.38
D512-q157	128/128/256	483	467	950	144.39	138.45	133.38
D1024-q251	256/256/256	1053	1021	2074	278.68	259.33	166.86
S512-q251 C139.5	256/256/256	543	511	1054	139.54	134.10	169.66
S512-q251 den128	256/256/256	543	511	1054	140.07	134.58	166.02
S1024-q349	256/512/256	1114	1082	2196	270.02	251.55	164.87
S1024-q347 den256	256/512/256	1113	1081	2194	270.02	251.54	163.66
S1024-q347 den512	256/512/256	1113	1081	2194	270.02	251.54	163.70
D1024-q269	256/256/256	1066	1034	2100	276.76	257.61	177.89
S512-q269 BAT M128	128/256/256	533	517	1050	144.01	138.13	149.20
S1024-q373 BAT	256/512/256	1126	1094	2220	272.02	253.35	170.66
S1024-q379 DFR256	256/512/256	1129	1097	2226	246.28	230.19	262.55
S1024-q347 DFR256	256/512/256	1113	1081	2194	192.29	181.59	271.09
S1024-q541 DFR256	256/512/256	1195	1163	2358	269.91	251.46	257.15
S1024-q347 C198.7	256/512/256	1113	1081	2194	199.07	187.70	269.42

M/u/K are external message / internal residue / shared-key output lengths in bits.
C/Q are supplied structural GSA/MATZOV diagnostics; b is approximate -log2(B), not a rounded strict inequality.
These columns are not a full KEM security claim or a NIST certificate. No compression denominator is assigned.

Table 4: Appendix: all16 family REF / AVX2 results

	REF (kcycles)			AVX2 (kcycles)		
Scheme	KeyGen	Encaps	Decaps	KeyGen	Encaps	Decaps
S512-q251	2,222.412 [2,220.931, 2,229.777]	300.676 [300.543, 300.744]	460.114 [459.999, 460.254]	1,843.785 [1,841.390, 1,845.584]	184.417 [184.247, 184.541]	232.698 [232.421, 232.929]
S1024-q347	4,917.572 [4,913.513, 4,920.746]	741.887 [741.716, 741.972]	1,065.250 [1,065.098, 1,065.653]	3,861.338 [3,859.300, 3,865.328]	484.334 [484.092, 484.443]	560.771 [560.662, 560.828]
D512-q157	2,260.666 [2,257.419, 2,264.095]	268.340 [268.166, 268.436]	427.895 [427.696, 428.057]	1,878.953 [1,875.310, 1,882.568]	152.781 [152.730, 152.863]	201.447 [201.392, 201.552]
D1024-q251	4,648.099 [4,642.902, 4,657.134]	696.215 [696.013, 696.385]	1,023.468 [1,023.243, 1,023.748]	3,568.026 [3,566.258, 3,571.190]	441.248 [441.100, 441.425]	521.723 [521.609, 521.894]
S512-q251 C139.5	2,236.392 [2,233.559, 2,243.607]	301.327 [301.190, 301.438]	460.782 [460.659, 460.920]	1,861.702 [1,860.086, 1,863.212]	185.047 [184.948, 185.145]	233.390 [233.192, 233.586]
S512-q251 den128	2,376.427 [2,368.728, 2,381.015]	301.203 [301.107, 301.365]	460.672 [460.586, 460.774]	1,999.910 [1,998.309, 2,001.831]	184.970 [184.859, 185.058]	233.068 [232.623, 233.408]
S1024-q349	4,684.743 [4,665.080, 4,689.082]	741.120 [740.893, 741.353]	1,066.891 [1,066.677, 1,067.222]	3,630.365 [3,619.905, 3,642.269]	484.459 [484.334, 484.727]	561.251 [561.134, 561.343]
S1024-q347 den256	4,651.786 [4,648.410, 4,661.042]	740.809 [740.582, 741.114]	1,064.078 [1,063.772, 1,064.346]	3,607.075 [3,601.068, 3,615.913]	483.371 [483.171, 483.534]	559.571 [559.503, 559.665]
S1024-q347 den512	4,605.953 [4,600.470, 4,617.949]	740.819 [740.669, 740.984]	1,064.010 [1,063.683, 1,064.297]	3,550.507 [3,547.322, 3,552.988]	483.127 [483.020, 483.238]	559.642 [559.533, 559.781]
D1024-q269	4,668.494 [4,650.249, 4,675.803]	704.758 [704.488, 704.905]	1,036.300 [1,036.079, 1,036.507]	3,567.649 [3,564.010, 3,569.429]	446.210 [446.124, 446.348]	526.776 [526.689, 526.865]
S512-q269 BAT M128	2,380.732 [2,377.540, 2,383.324]	301.245 [301.186, 301.340]	459.749 [459.620, 459.888]	2,003.691 [1,997.234, 2,007.713]	185.544 [185.476, 185.618]	234.055 [234.007, 234.147]
S1024-q373 BAT	4,913.830 [4,910.288, 4,915.833]	747.292 [747.052, 747.478]	1,074.104 [1,073.967, 1,074.487]	4,034.410 [4,030.482, 4,039.471]	486.595 [486.502, 486.737]	563.431 [563.225, 563.538]
S1024-q379 DFR256	4,894.446 [4,778.300, 4,903.992]	743.559 [743.485, 743.820]	1,065.767 [1,065.592, 1,066.022]	3,704.537 [3,685.873, 3,707.675]	487.717 [487.664, 487.782]	564.199 [564.079, 564.300]
S1024-q347 DFR256	4,608.497 [4,601.757, 4,614.139]	740.842 [740.669, 741.063]	1,063.851 [1,063.598, 1,064.146]	3,552.851 [3,549.360, 3,557.075]	483.012 [482.929, 483.210]	559.485 [559.383, 559.590]
S1024-q541 DFR256	5,523.443 [5,509.782, 5,575.542]	824.696 [824.582, 824.806]	1,148.908 [1,148.421, 1,149.220]	4,406.209 [4,403.880, 4,412.590]	561.937 [561.794, 562.100]	635.021 [634.926, 635.195]
S1024-q347 C198.7	4,601.489 [4,597.831, 4,608.790]	740.877 [740.693, 741.162]	1,064.020 [1,063.604, 1,064.347]	3,552.322 [3,549.014, 3,556.473]	483.263 [483.028, 483.373]	559.655 [559.551, 559.723]

Median of accepted batch means; brackets give the prespecified bootstrap 95% interval.
1 kcycle = 1000 raw hardware CPU cycles. Counter-read overhead is retained, not subtracted.
Blank performance cells are correctness exclusions, never zero costs. Batch counts are retained in CSV.

Table 5: Matched peer API wire sizes

Scheme	M/u/K (bits)	API CT (bytes)	API PK (bytes)	API total (bytes)	Source CT (bytes)	Source PK (bytes)
END-512	256/—/256	384	514	898	384	514
BAT-512	128/—/—	473	521	994	472	520
END-1024	512/—/256	832	1027	1859	832	1027
BAT-1024	256/—/—	1006	1230	2236	1005	1229

API sizes come from the matched build descriptors; source sizes retain the archive convention.
BAT API keys and ciphertexts include one-byte headers beyond the header-free source metadata.
— means the source does not specify that length; message length does not imply shared-key length.

Table 6: Published comparator metadata (separate evidence models)

Scheme	M/u/K (bits)	Source CT (bytes)	Source PK (bytes)	C reported	Q reported	DFR exp. (reported)
DAWN-alpha-512	128/128/—	436	615	139.50	133.50	133.00
DAWN-alpha-512-prime	256/256/256	514	615	140.00	—	166.00
DAWN-alpha-1024	256/256/—	973	1229	269.90	250.80	163.00
BAT-512	128/—/—	472	520	140.30	124.70	146.70
BAT-1024	256/—/—	1005	1229	283.20	250.60	166.70
END-512	256/—/256	384	514	152.70	—	130.00
END-1024	512/—/256	832	1027	287.10	—	157.00
DAWN-beta-512	128/128/—	450	514	134.30	134.10	130.00
DAWN-beta-1024	256/256/—	1027	1027	266.50	251.20	138.00

Source-only sizes and model/DFR statements; no illustration timing is used.
Published C/Q and DFR statements use heterogeneous models and must not be ranked as a common security scale.
Exact attribution, relations, qualifications and URLs are retained in metadata JSON/CSV and Markdown.